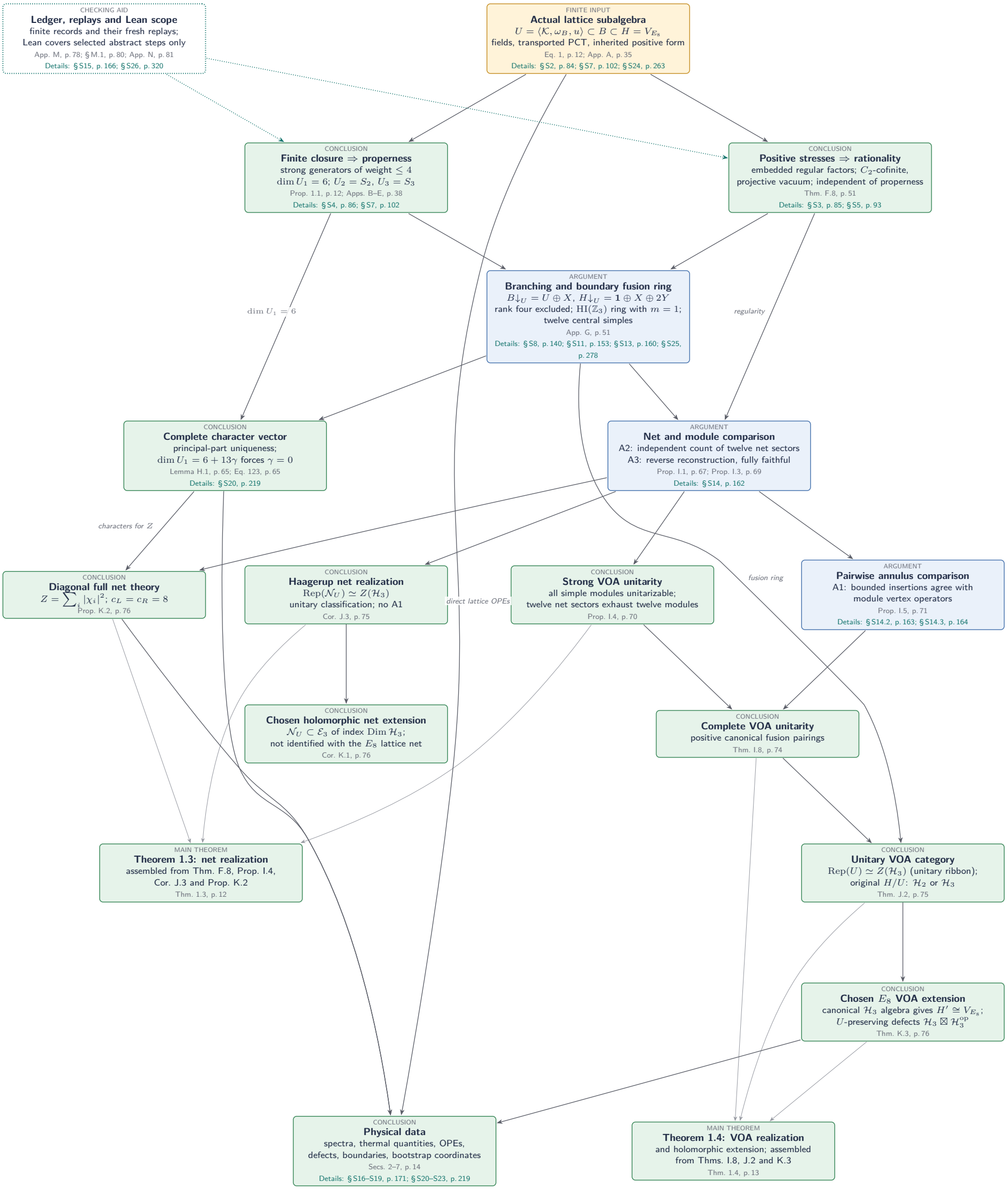
Claude

Argument overview

The construction, the net realization, the VOA realization and the observables have different dependencies. Teal lines name the supplementary appendices that print the supporting data.

Finite input | Argument in draft | Conclusion | Checking aid



Arrows show selected logical dependencies, not chronology. Incoming arrows are used jointly; transitive dependencies are usually omitted. Thin arrows show how the two main theorems are assembled. Pages 2–7 resolve each block into its steps.

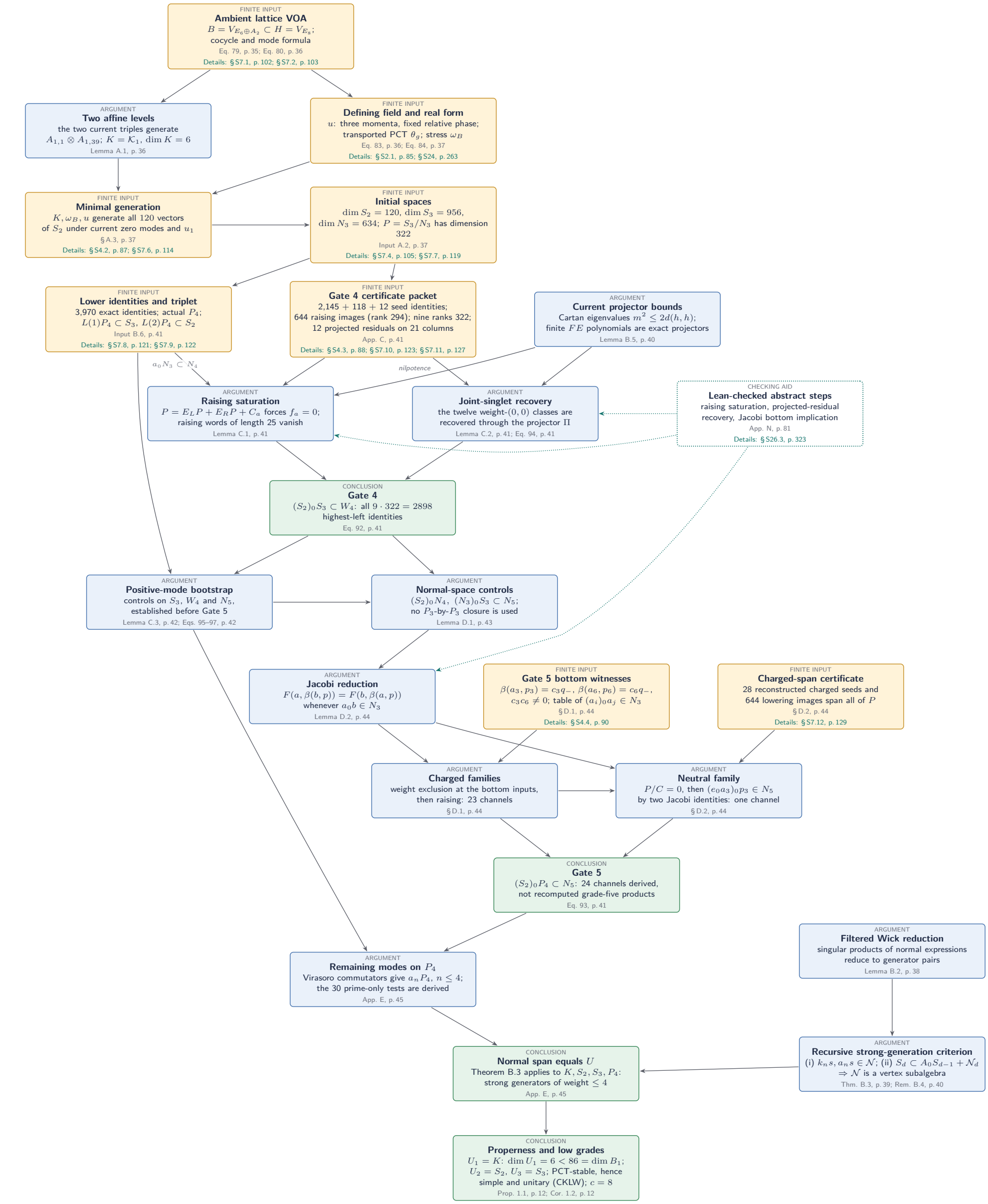
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Finite construction and the all-weight argument

Appendices A–E: actual-vector provenance comes before closure, the two gates are derived from finite witnesses, and properness comes before any character identification.

Finite input | Argument in draft | Conclusion | Checking aid



The projected residuals alone do not prove Gate 4: raising premises and saturation ranks are also used. Gate 5 and its positive-mode consequences are derived; the value $\dim U_4 = 6010$ is not an input. Every finite input is listed with its record in the certificate ledger (App. M) and its fresh replay in §M.1.

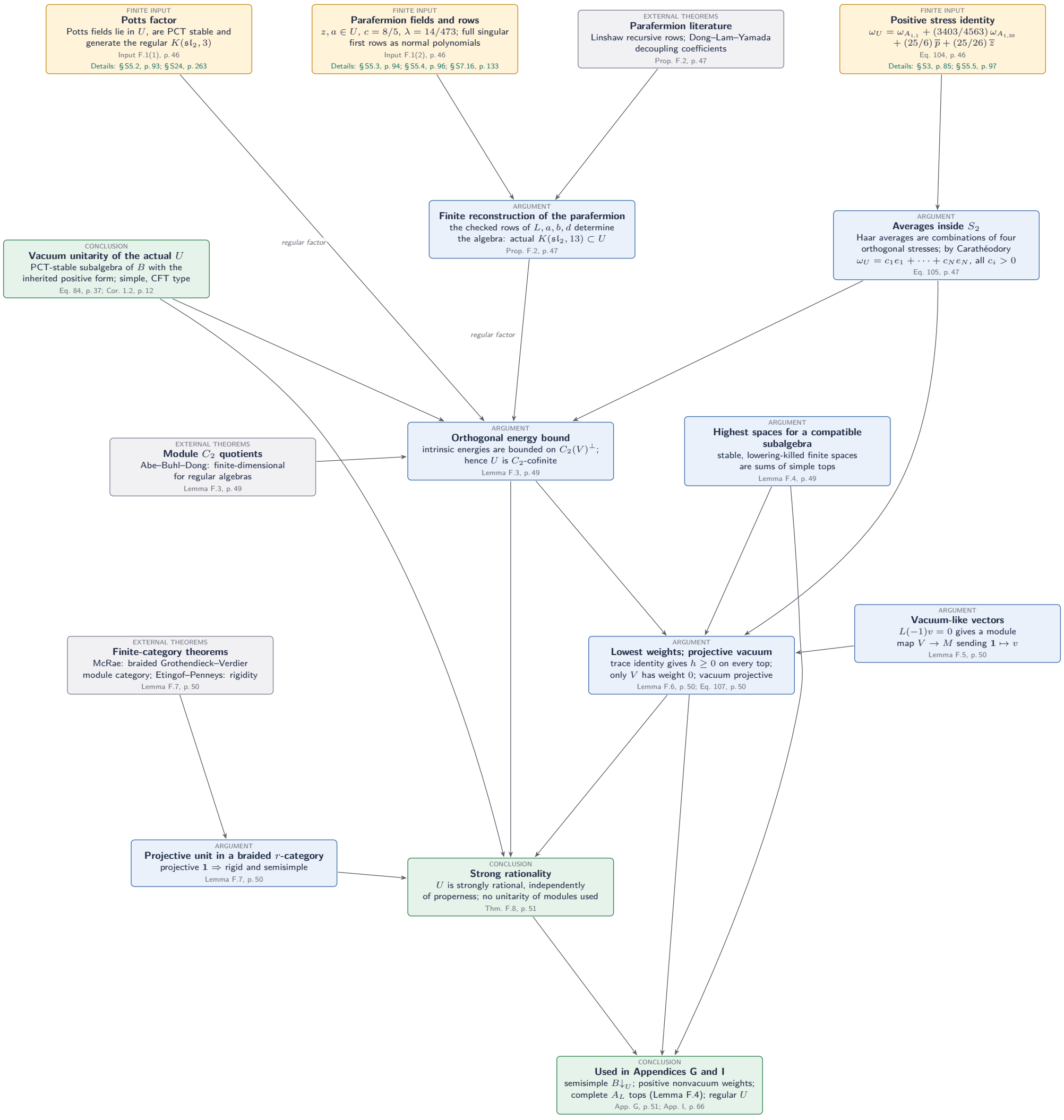
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Regularity and strong rationality

Appendix F works with the actual PCT-stable algebra $U \subset B$. None of its conclusions uses properness or the finite closure gates.

Finite input | Argument in draft | Conclusion | External input



The embedded regular factors and the positive stress identity are exact field inputs (review note R1 of App. F); Proposition F.2 transports the checked parafermion rows to an identification but does not replace their verification. Lemma F.7 does not assume that nonvacuum modules are unitarizable.

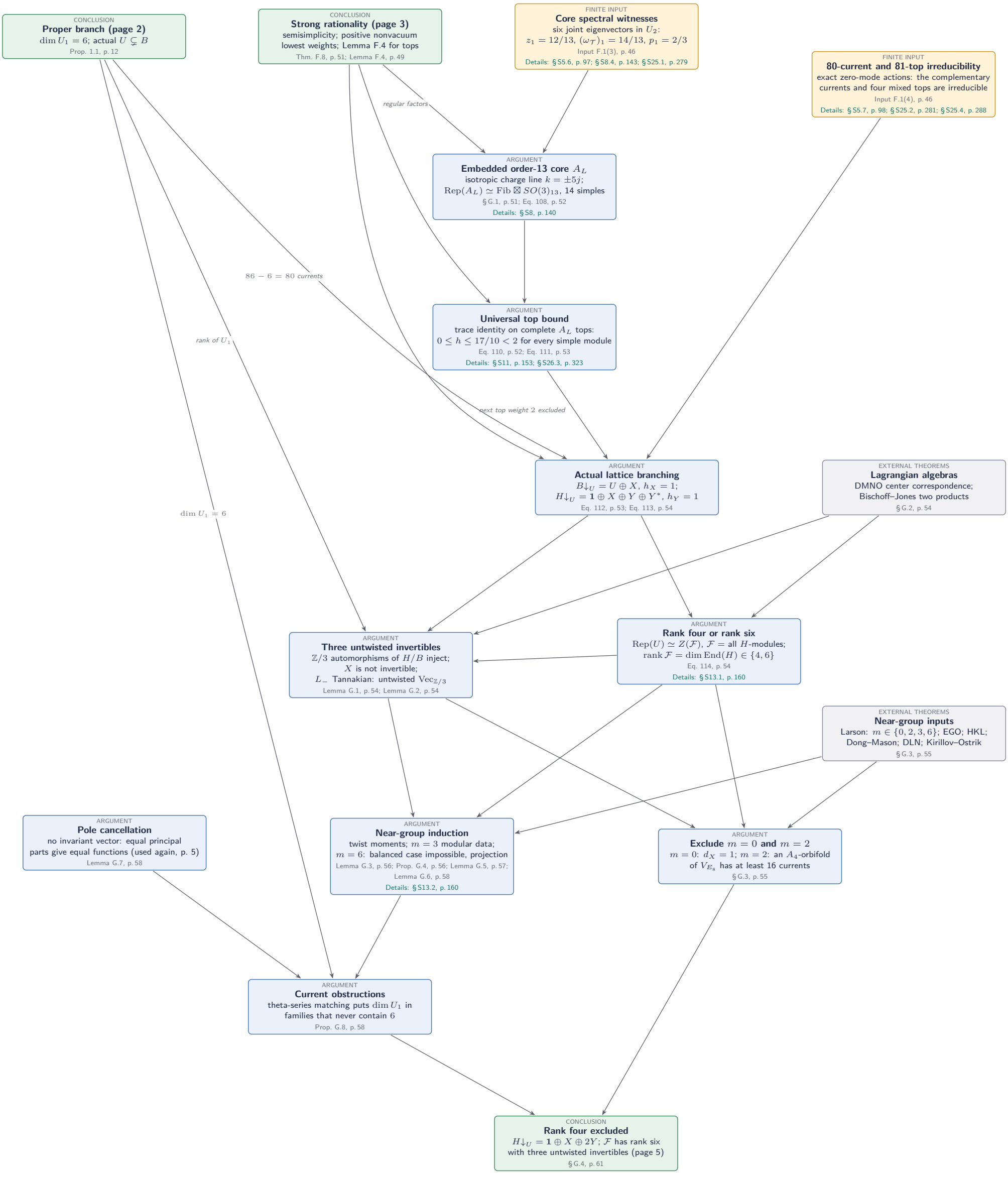
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Branching and the rank fork

Appendix G, first half: the actual lattice branching fixes the boundary category up to rank four or six, and rank four is excluded before any unitary classification is invoked.

Finite input | Argument in draft | Conclusion | External input



Larson's near-group theorem assumes pseudo-unitarity; for $\mathcal{F}$ it follows from that of its center, whose positive spherical structure comes from strictly positive nonvacuum lowest weights. No C^* structure is used here. Lemma G.2 replaces the earlier pointwise stabilizer computation $\mathrm{Aut}(B/U) = 1$.

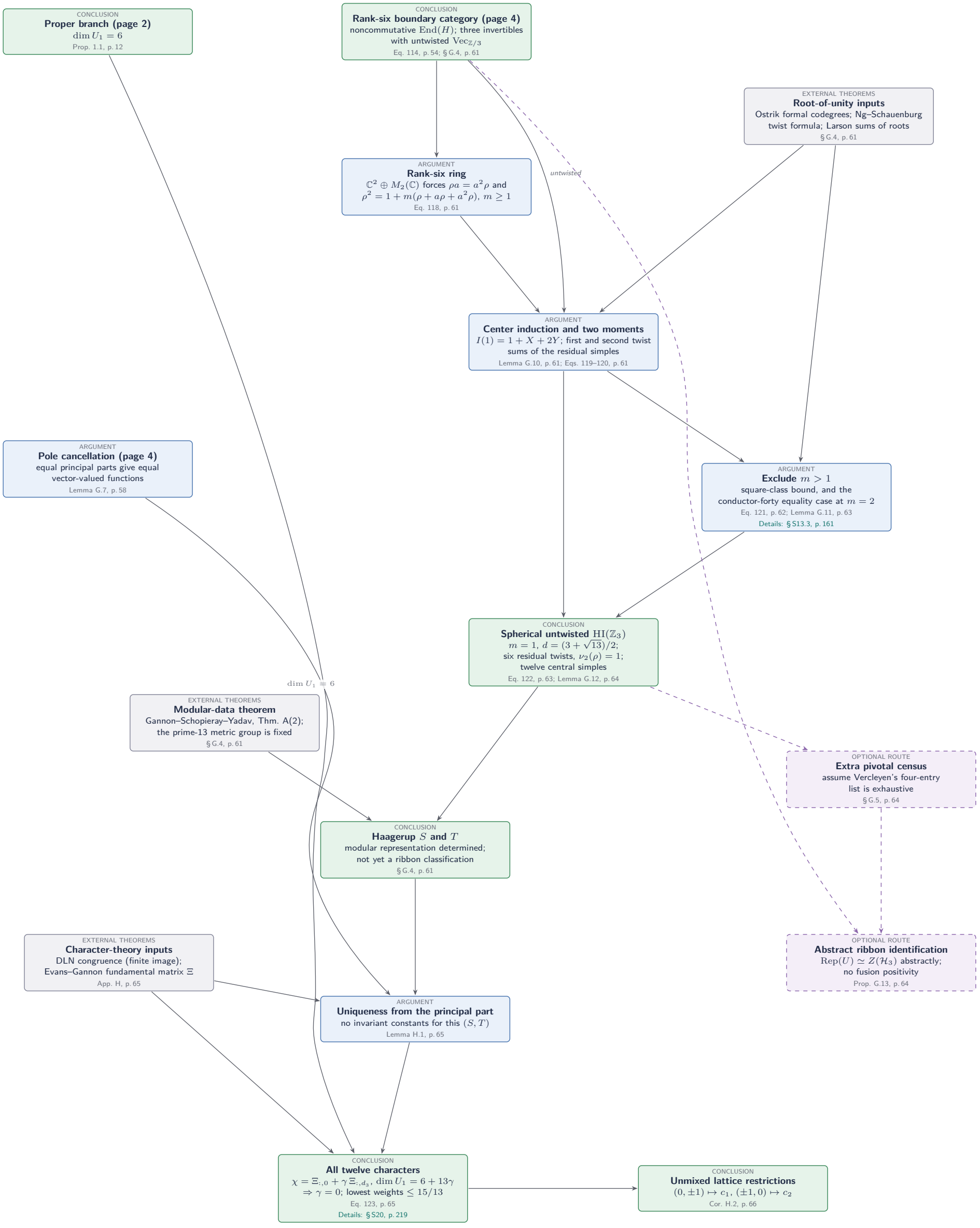
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Rank six, modular data and the characters

Appendix G, second half, and Appendix H: the character proof uses modular data derived from the actual branching, not an assumed tensor-category identification.

Argument in draft | Conclusion | External input | Optional route



Dashed arrows mark an optional route using an extra pivotal-classification hypothesis. It is not used for the characters or the analytic realization, and it does not establish canonical VOA fusion positivity. The character vector fixes every graded dimension but was not used to prove properness.

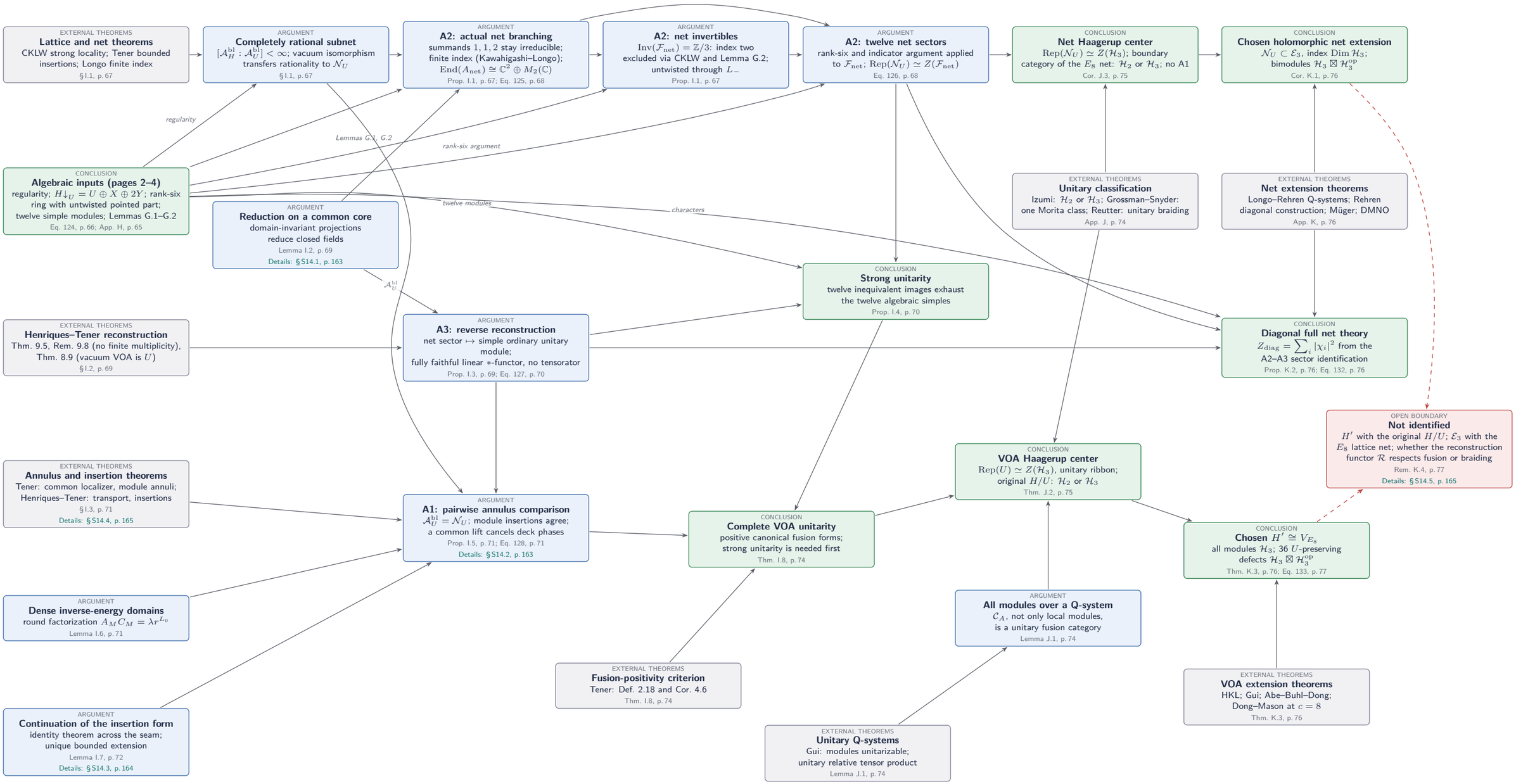
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Analytic realization: two distinct routes

Appendices I–K: positivity of the vacuum, of all modules and of the fusion pairings are separate assertions; A2–A3 and A1 are different bridges.

Argument in draft | Conclusion | External input | Open boundary



A2 and A3 share core methods but are parallel arguments. The net route needs no A1; the chosen holomorphic net is not identified with E_8. The chosen VOA extension H' need not equal the original embedding H/U. Haagerup categories use all algebra modules, not only local modules.

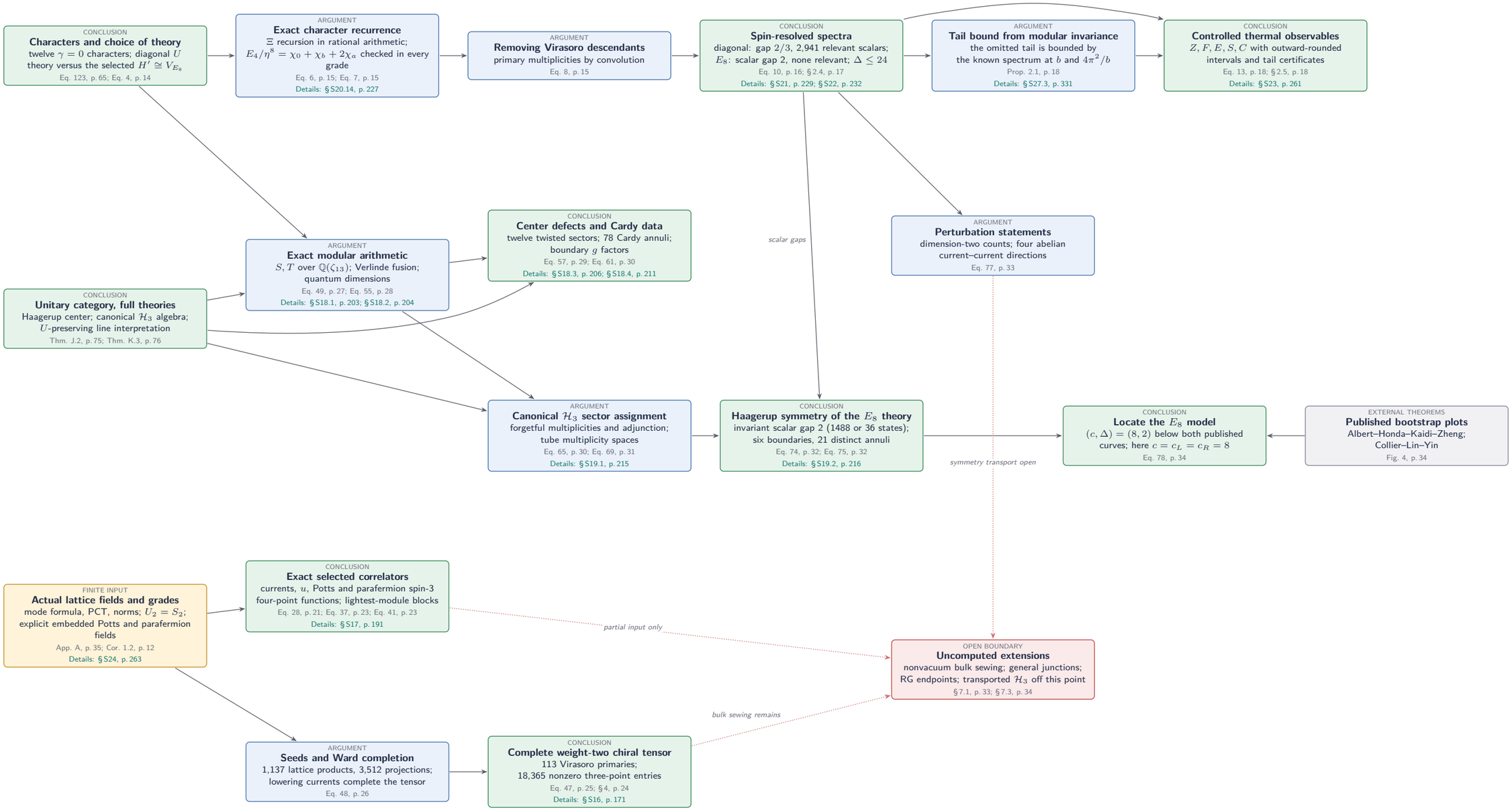
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From the construction to physical observables

Sections 2–7: each dataset has explicit mathematical inputs, and the two full theories are kept apart.

Finite input | Argument in draft | Conclusion | External input | Open boundary



The bootstrap point uses $c = c_L = c_R = 8$, not 16. Diagram endpoints record what the draft computes; they do not assert a complete bulk OPE, a new certified SDP exclusion bound, or a symmetry-preserving conformal manifold.

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Where the supplementary appendices attach

The supplementary appendices S1–S28 print the records behind the nodes. Supplementary routes marked "–" carry their own hypotheses and are not used by the main argument.

Part	Page	Contents	Used for	Flowchart page
§ S1	83	How to audit the inputs	reading order for the finite inputs	2–5
§ S2	84	The small lattice calculation	defining field, levels, coefficient field	2
§ S3	85	Four-column check of the positive stress identity	Eq. 104	3
§ S4	86	Finite closure identities a reader can inspect	minimal generation, Gate 4 residuals, Gate 5 scalars	2
§ S5	93	Structural inputs for direct checking	Potts and parafermion rows, membership, core witnesses, irreducibility	3–4
§ S6	101	Reproduction and the remaining large checks	scope of the small checker and replays	2–5
§ S7	102	Expanded finite-lattice reference	coordinates, generation graph, grade three, 3,970 identities, Gate 4 and Gate 5 records	2–3
§ S8	140	Atlas of the commuting algebras	lattice–Potts core and its order-13 extension A_L	4
§ S9	144	Residual conformal vectors	supplementary; not used by the main argument	–
§ S10	149	Ambient residual cosets	supplementary route; not needed for regularity or closure	–
§ S11	153	Finite top data behind the energy bound	universal top bound $h \leq 17/10$	4
§ S12	157	Charge-six lattice coset	supplementary route with its own hypotheses	–
§ S13	160	Categorical exclusions: diagnostics	rank fork, multiplicity six, rank-six check	4–5
§ S14	162	Analytic verification map	A2, A3, A1; the three extension objects	6
§ S15	166	Structural provenance and replays	replay status of the structural inputs	3–4
§ S16	171	Weight-two interaction atlas	complete weight-two chiral tensor	7
§ S17	191	Correlator tables and channel expansions	exact selected correlators	7
§ S18	203	Modular, fusion and diagonal-defect tables	modular arithmetic, defects, Cardy data	7
§ S19	214	Six Haagerup lines and extension boundaries	$\mathcal{H}_3$ sector assignment and E_8 symmetry	7
§ S20	219	Characters through grade 24	character vector and recurrence	5, 7
§ S21	229	Full scalar spectrum	spin-resolved spectra	7
§ S22	232	Spin-resolved spectrum	spin-resolved spectra	7
§ S23	261	Thermal intervals and tail certificates	controlled thermal observables	7
§ S24	263	Coordinate atlas of the small fields	defining field, Potts and parafermion fields	2–3, 7
§ S25	278	Finite top witness atlas	core witnesses; 80-current and 81-top irreducibility	4
§ S26	320	Lean formalization: coverage	abstract steps of Apps. C, D and G; thermal tail algebra	2, 4, 7
§ S27	329	Assessment of the physics revision	scope of each physics dataset	7
§ S28	335	Source concordance and assembly	result-to-source map for every page	all

Appendices A–O keep the numbering of the 78-page base revision of 27 September 2026; only their pages moved. The verification guide (S1–S6) and the formalization assessment (S26–S27) are incorporated companions whose references now resolve inside the expanded draft.

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